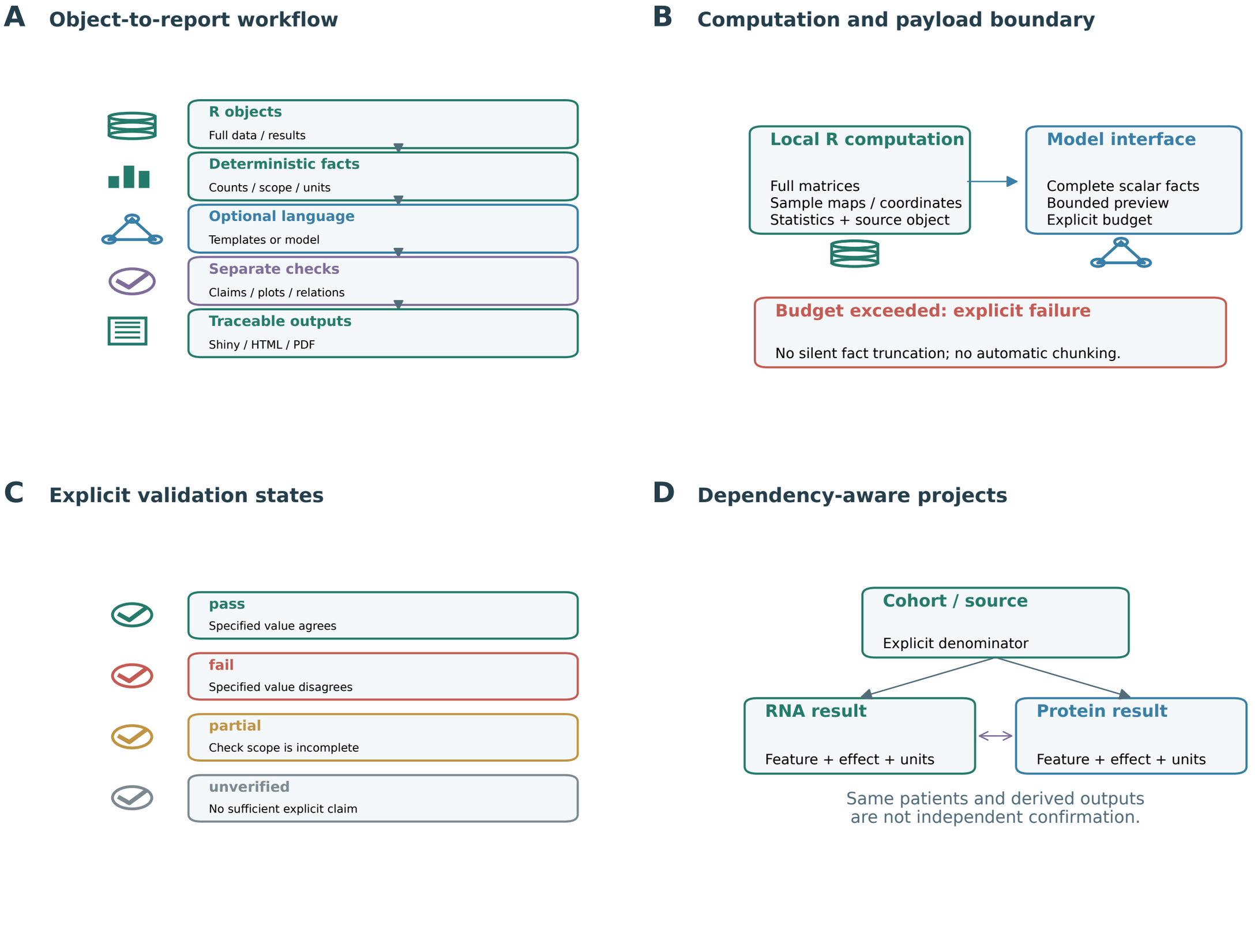

A
Object-to-report workflow
R objects
Full data / results
Deterministic facts
Counts / scope / units
Optional language
Templates or model
Separate checks
Claims / plots / relations
Traceable outputs
Shiny / HTML / PDF
B
Computation and payload boundary
Local R computation
Full matrices
Sample maps / coordinates
Statistics + source object
Model interface
Complete scalar facts
Bounded preview
Explicit budget
Budget exceeded: explicit failure
No silent fact truncation; no automatic chunking.
C
Explicit validation states
pass
Specified value agrees
fail
Specified value disagrees
partial
Check scope is incomplete
unverified
No sufficient explicit claim
D
Dependency-aware projects
Cohort / source
Explicit denominator
RNA result
Feature + effect + units
Protein result
Feature + effect + units
Same patients and derived outputs
are not independent confirmation.